



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category - I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

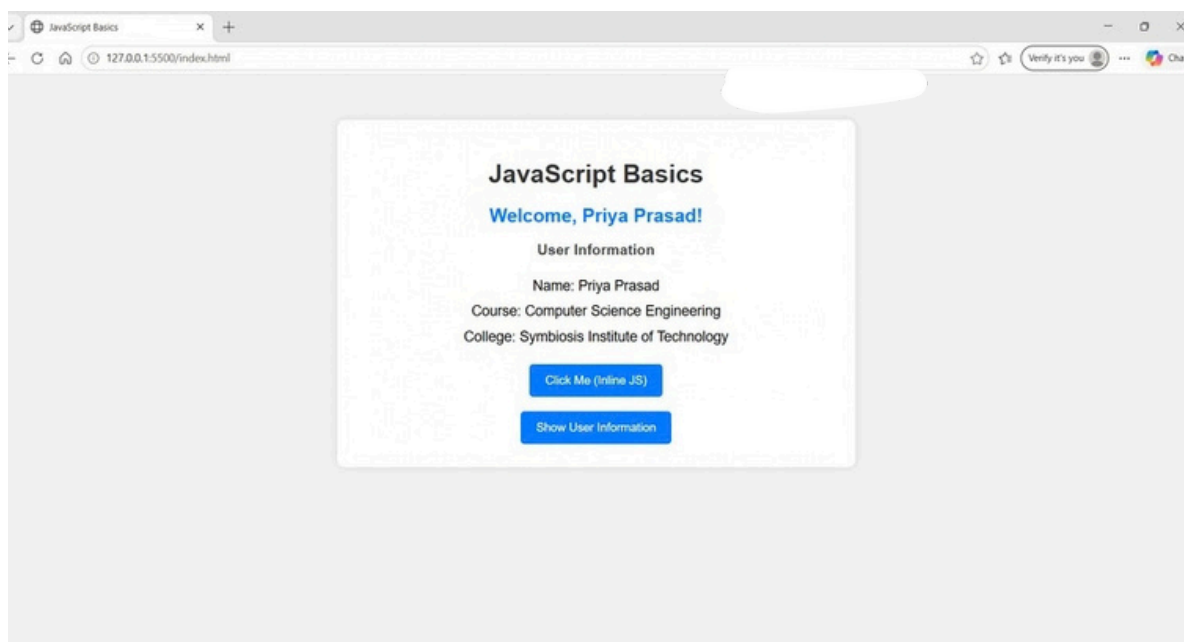
Experiment No: 01

1. Title: Demonstrate inline, internal, and external JavaScript; use console methods; create a webpage showing user info and a welcome message.

2. Software/Tools Required: HTML, VS Code.

3. Theory: HTML (HyperText Markup Language) is the standard language used to create web pages. It provides the basic structure of a webpage using elements such as headings, paragraphs, images, lists, tables, and buttons. CSS (Cascading Style Sheets) is used to style the webpage. It controls the layout, colors, fonts, spacing, and overall appearance of the webpage. In this program, an "About Me" webpage is created using HTML and CSS. The webpage contains a profile picture, personal information, skills, education details, and a contact button. CSS is used to improve the appearance by adding colors, spacing, borders, and alignment.

4. Output:





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5.Conclusion: In this assignment, we successfully demonstrated inline, internal, and external JavaScript. We learned how to use console methods, display user information, and show a welcome message on a webpage. This assignment helped us understand the basics of JavaScript and its interaction with HTML using DOM manipulation.



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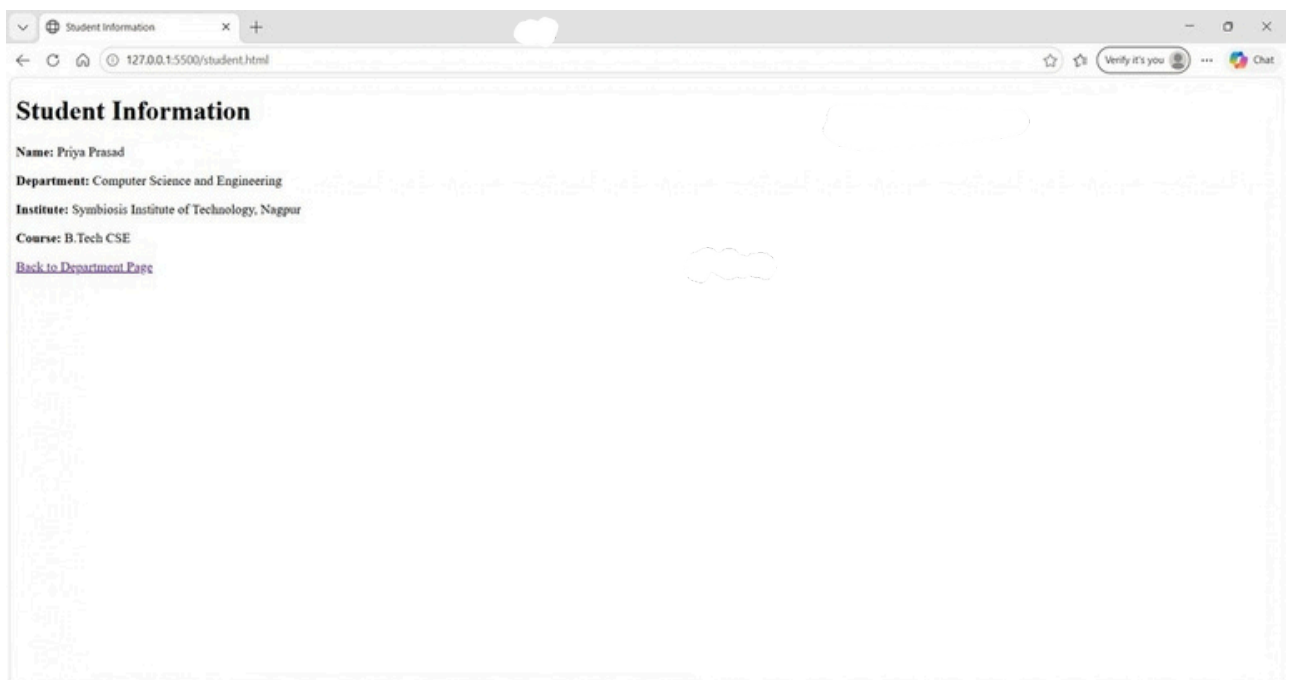
Case Study : 1

1. Title: Create a Webpage for College Department and Link the Student Information Page Using JavaScript Alert.

2. Software/Tools Required: HTML, VS Code.

3. Theory: This case study focuses on creating a college department webpage using HTML and JavaScript. The webpage displays information about the Computer Science and Engineering Department of SIT Nagpur. A separate student information page is linked to the department webpage using an HTML hyperlink. JavaScript is used to display a welcome message, "Welcome to SIT nagpur", in an alert box when the user clicks the Welcome button

4. Output:





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Conclusion : In this case study, we successfully created a college department webpage and linked it to a student information page. We also used JavaScript to display a welcome message in an alert box. This activity helped us understand HTML page linking and basic JavaScript alert functions.



Experiment No.: 2

1. **Title:** Billing Calculator Using JavaScript ES6 Features

2. **Software/Tools Required:** Visual Studio Code (VS Code), web browser, HTML, JavaScript.

3. **Theory:** JavaScript provides different features to make programming easier and more efficient. var, let, and const are used to declare variables. Template literals allow us to display dynamic values using `\${}`. Destructuring is used to extract values from objects. User input is taken through HTML input fields, and JavaScript performs calculations based on the entered price and quantity. In this experiment, a billing calculator is developed to calculate the total amount, tax, and final bill using these JavaScript feature

Output :

Billing Calculator

Pencil Box

50

2

Calculate Bill

My Shop

Item: Pencil Box

Price: ₹50

Quantity: 2

Total: ₹100.00

Tax (5%): ₹5.00

Final Amount: ₹105.00

Conclusion: The billing calculator was successfully implemented using JavaScript. The experiment demonstrated the use of var, let, const, template literals, destructuring, and user input. The program calculates the total bill and displays the final amount dynamically.



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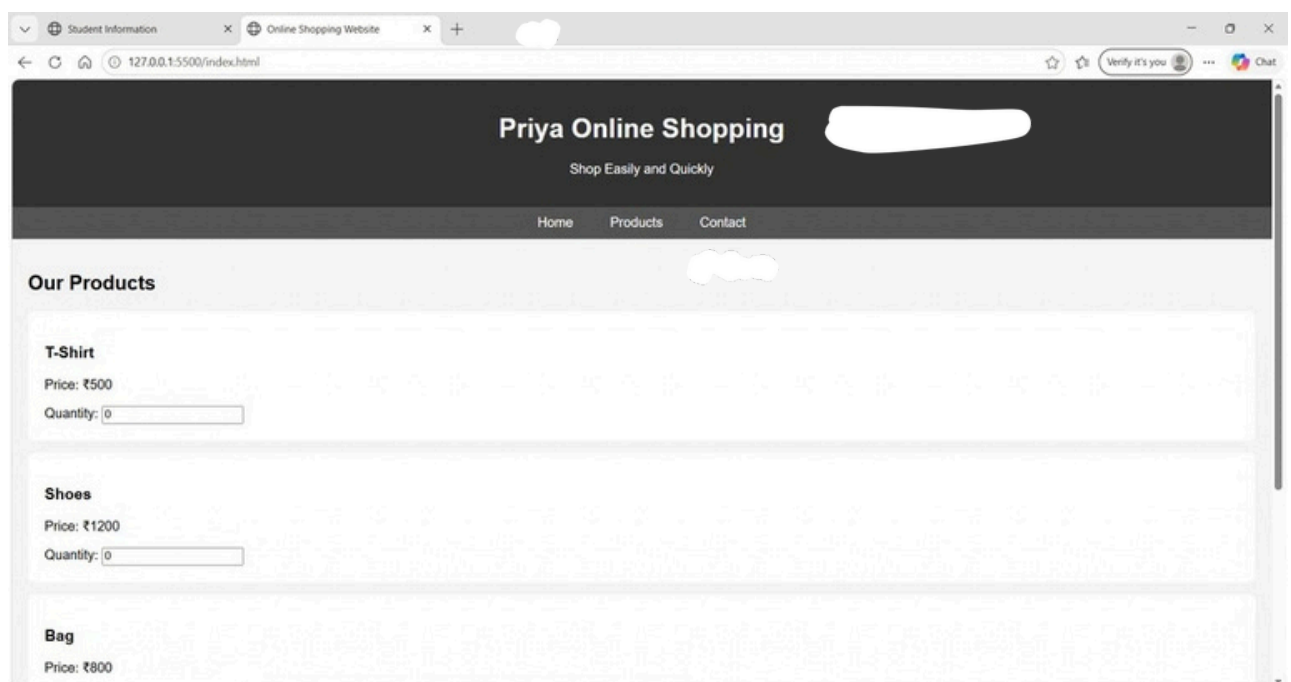
Case Study : 2

1. Title: Develop an Online Shopping Website Using HTML, CSS, HTML5 Tags, and JavaScript to Calculate the Bill.

2. Software/Tools Required: HTML, VS Code.

3. Theory: This case study focuses on developing an online shopping website using HTML, CSS, and JavaScript. HTML5 semantic tags such as `<header>`, `<nav>`, `<main>`, `<section>`, `<article>`, and `<footer>` are used to structure the webpage. CSS is used to design and style the website. JavaScript is used to accept product quantities, calculate the total price, add GST, and display the final bill dynamically on the webpage.

4. Output:





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Conclusion : In this case study, we successfully developed an online shopping webpage using HTML, CSS, and JavaScript. We learned how to use HTML5 semantic tags, apply CSS styling, and calculate the total bill using JavaScript. The webpage displays product details and generates the final bill based on the selected quantities.



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Experiment No: 3

1. Title: Implementation of Control Structures and Form Validation Using JavaScript

2. Software/Tools Required: HTML, VS Code.

3. Theory: JavaScript control structures such as if, else if, and else are used to make decisions based on user input. In this assignment, a grading system is developed to calculate a student's grade based on their marks. Form validation is used to check whether the user has entered valid information. The program validates the student name and marks (between 0 and 100). Based on the marks, an appropriate grade is assigned using conditional statements. Event handling is used to process the form submission and display the result dynamically on the webpage.

4. Output:

Student Grading System

Student Name:

Enter Marks (0-100):

Student Name: Priya Prasad
Marks: 75
Grade: B
Good Performance



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Conclusion: The student grading system was successfully implemented using JavaScript. The assignment helped in understanding control structures, form validation, event handling, and user input processing. The program calculates grades based on marks and displays the student's result dynamically.



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Case Study : 3

1. Title: Scholarship Eligibility Checker Using JavaScript

2. Software/Tools Required: HTML, VS Code.

3. Theory: The Scholarship Eligibility Checker is a web-based application developed using HTML, CSS, and JavaScript. It takes the student's marks and annual family income as input. JavaScript uses conditional statements (if-else) to check whether the student meets the given eligibility criteria. The result is displayed dynamically on the webpage using DOM manipulation.

4. Output:



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Conclusion : In this case study, we successfully developed a Scholarship Eligibility Checker using JavaScript. The application checks the student's marks and family income using conditional statements and displays the eligibility result. This activity helped us understand JavaScript conditions, user input, and DOM manipulation.



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Experiment No: 4

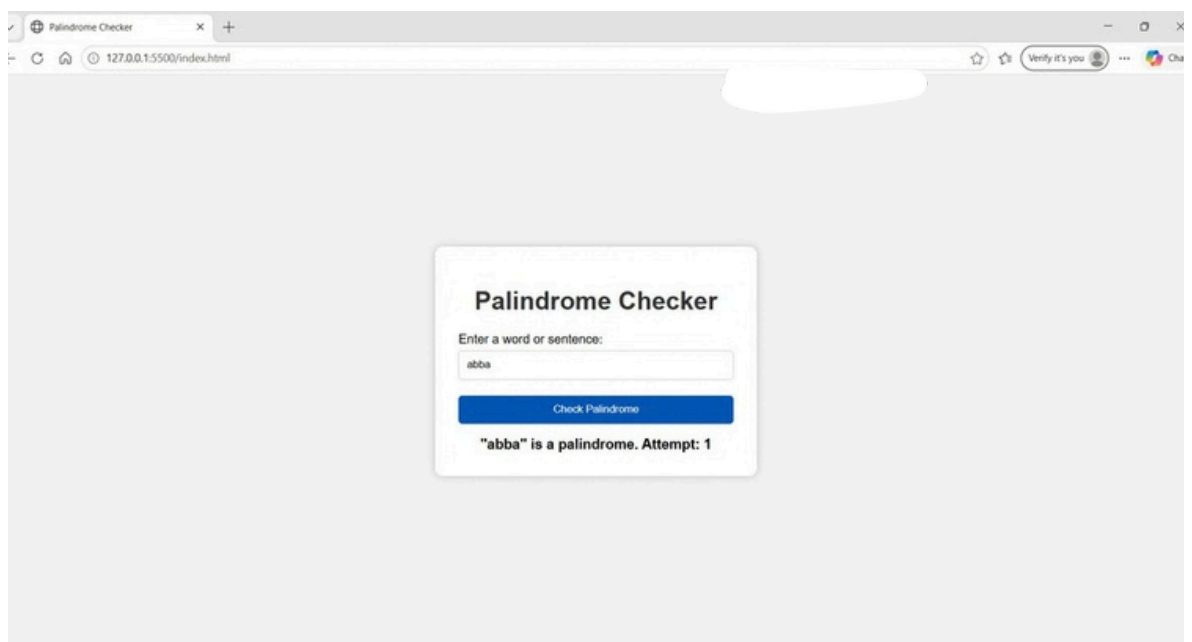
1. Title: Implementation of Functions, Scope, Closures, and Error Handling Using JavaScript

2. Software/Tools Required: HTML, VS Code.

3. Theory: JavaScript functions are reusable blocks of code that perform specific tasks. In this assignment, different types of functions, such as function declarations, function expressions, and arrow functions, are used. Scope defines the accessibility of variables, while closures allow a function to remember variables from its outer scope.

The try-catch statement is used for error handling and helps manage errors without stopping the entire program. A palindrome checker is developed to determine whether a word or sentence reads the same forward and backward. The result is displayed dynamically on the webpage.

4. Output:





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Conclusion: In this assignment, we successfully implemented a palindrome checker using JavaScript. We learned about different types of functions, variable scope, closures, and error handling using try-catch. The program checks user input and displays the result dynamically, improving our understanding of JavaScript programming concepts.



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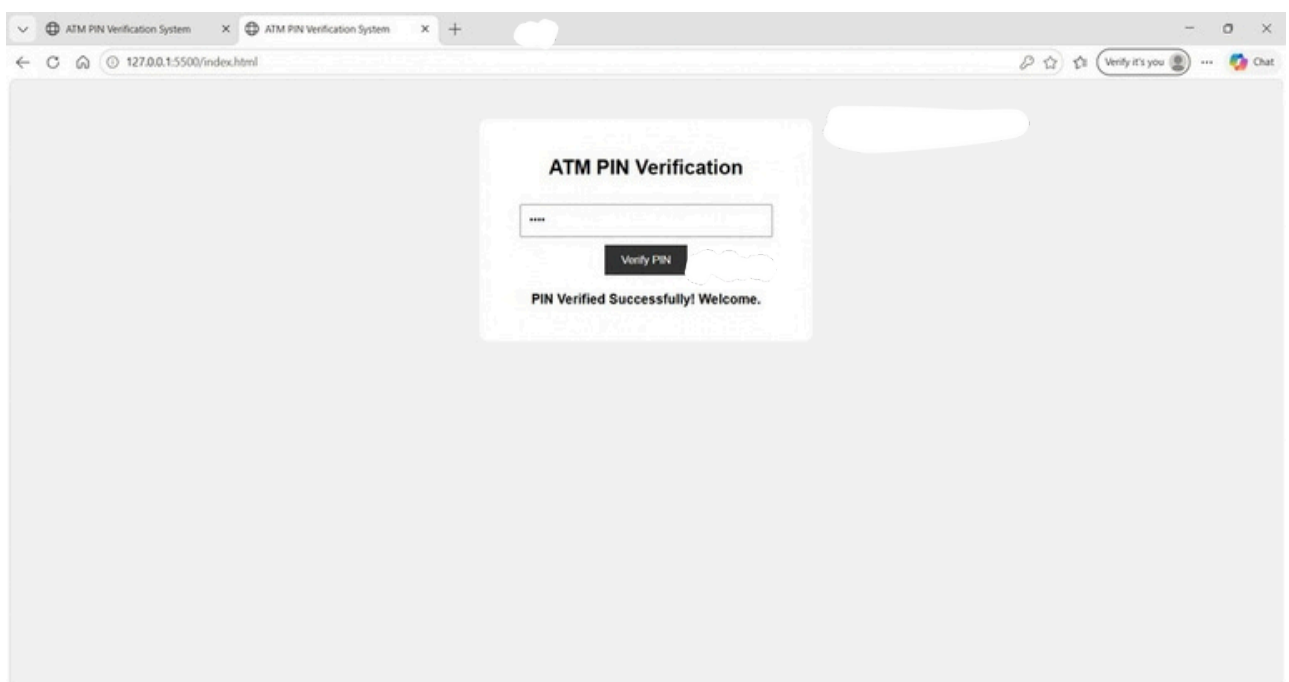
Case Study : 4

1. Title: ATM Card PIN Verification System Using JavaScript Functions, Closures, and Exception Handling

2. Software/Tools Required: HTML, VS Code.

3. Theory: The ATM Card PIN Verification System is developed using HTML, CSS, and JavaScript. It verifies the PIN entered by the user and checks whether it is correct. The program demonstrates different JavaScript concepts, including function declaration, function expression, arrow functions, variable scope, and closures. Exception handling using try-catch is used to handle invalid PIN input and maximum login attempts. The system displays appropriate messages based on the verification result.

4. Output:





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Conclusion : In this case study, we successfully implemented an ATM Card PIN Verification System using JavaScript. We learned how to use different types of functions, variable scope, closures, and exception handling. The program validates the entered PIN and displays suitable messages for successful and unsuccessful verification. This activity helped us understand the practical use of JavaScript functions and error handling.

Experiment No.: 5

1. **Title:** Apply array methods and object handling; create a cart total calculator with discount logic.

2. **Software/Tools Required:** Visual Studio Code (VS Code), web browser, HTML, JavaScript.

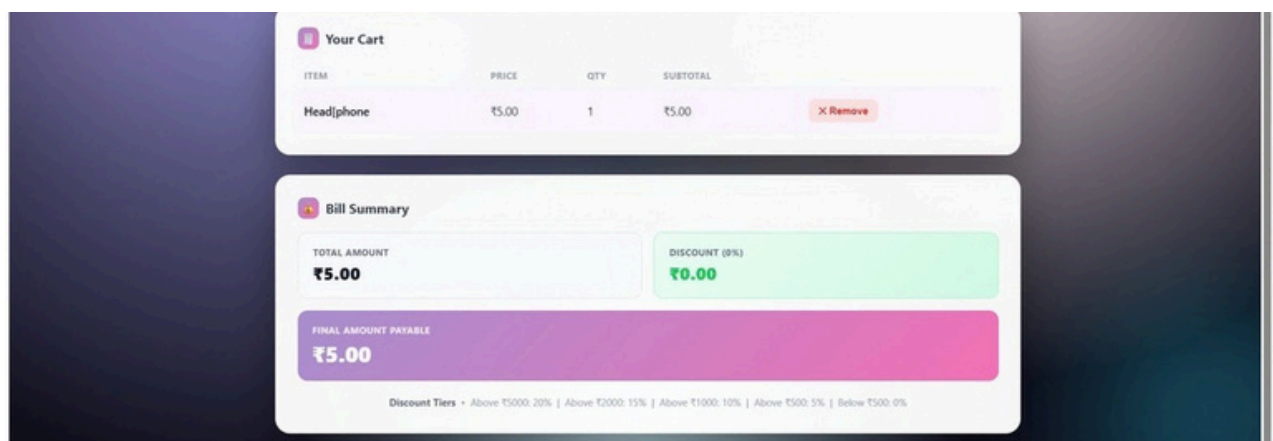
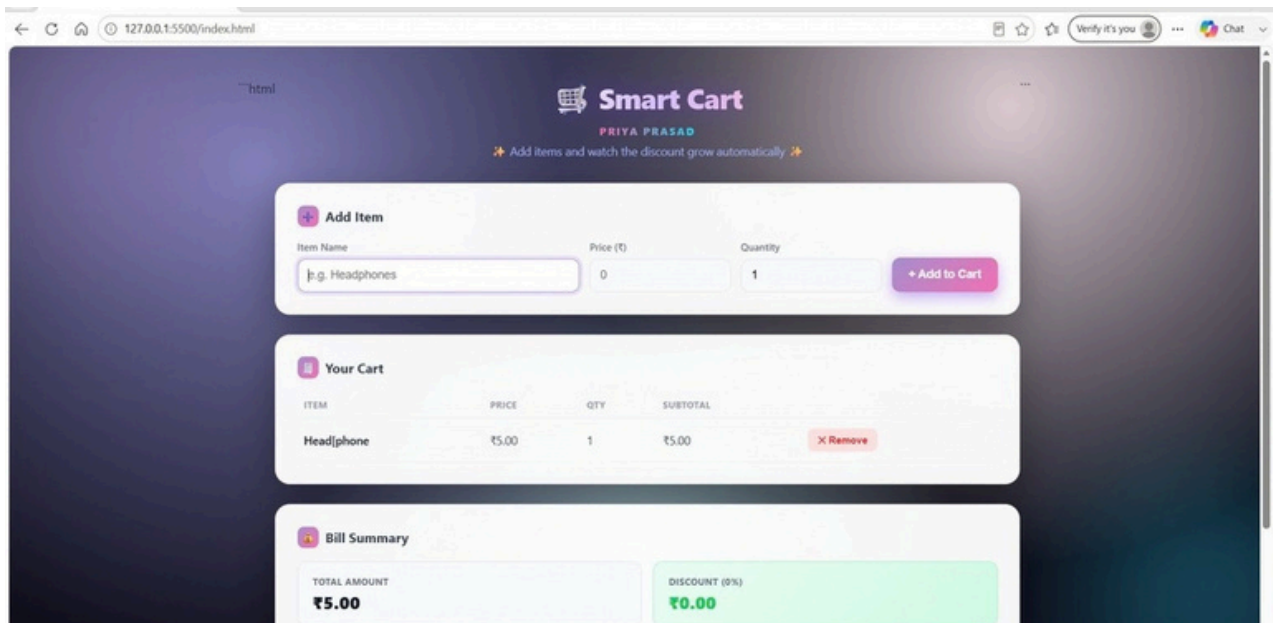
3. **Theory:** A Smart Cart Total Calculator is a JavaScript-based application that calculates the total price of items added to a shopping cart. It uses arrays to store cart items and objects to store details such as item name, price, and quantity. JavaScript array methods like `map()`, `reduce()`, and `filter()` are used to process cart data efficiently. Object handling is used to access and update product information. After calculating the subtotal, discount logic is applied based on predefined conditions, and the final payable amount is displayed.

Main concepts used:

- Arrays: Store multiple cart items.
- Objects: Store product details.
- `map()`: Process or modify item data.
- `reduce()`: Calculate the total cart value.
- `filter()`: Select items based on conditions.
- Conditional statements: Apply discounts.
- JavaScript functions: Organize the calculation logic.

This project helps understand how JavaScript handles arrays, objects, calculations, and conditional discount logic in a practical shopping-cart application.

Output :



Conclusion: The Smart Cart Total Calculator successfully demonstrates the use of JavaScript arrays, objects, and array methods in a practical application. It calculates item subtotals and the total cart value using `forEach()` and `reduce()`. The discount logic automatically applies different discounts based on the total amount, and the final payable amount is displayed clearly. This project provides a simple understanding of how JavaScript can be used to create an interactive shopping cart with dynamic calculations.



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Case Study : 5

1. Title: Study of Arrays and Loops in JavaScript with Maximum and Minimum Value Calculation.

2. Software/Tools Required: HTML, VS Code.

3. Theory: Arrays in JavaScript are used to store multiple values in a single variable. Array methods such as `push()`, `pop()`, `shift()`, `unshift()`, `splice()`, and `slice()` are used to add, remove, and extract elements from an array. Methods like `map()`, `filter()`, `reduce()`, and `forEach()` help in processing and manipulating array elements. In this case study, an array of numbers is created, and JavaScript is used to find the maximum and minimum values using `Math.max()` and `Math.min()`.

4. Output:

```
Arrays and Loops in JavaScript
Original Array: 10,20,30,40,50
After push(60): 10,20,30,40,50,60
After pop(): 10,20,30,40,50
After unshift(5): 5,10,20,30,40,50
After shift(): 10,20,30,40,50
After splice(): 10,20,35,40,50
After slice(1,4): 20,35,40

forEach(): 10 20 35 40 50
map() - Doubled Array: 20,40,70,80,100
filter() - Numbers > 20: 35,40,50
reduce() - Sum: 155

Case Study
Array: 10,20,35,40,50
Maximum Value: 50
Minimum Value: 10
```



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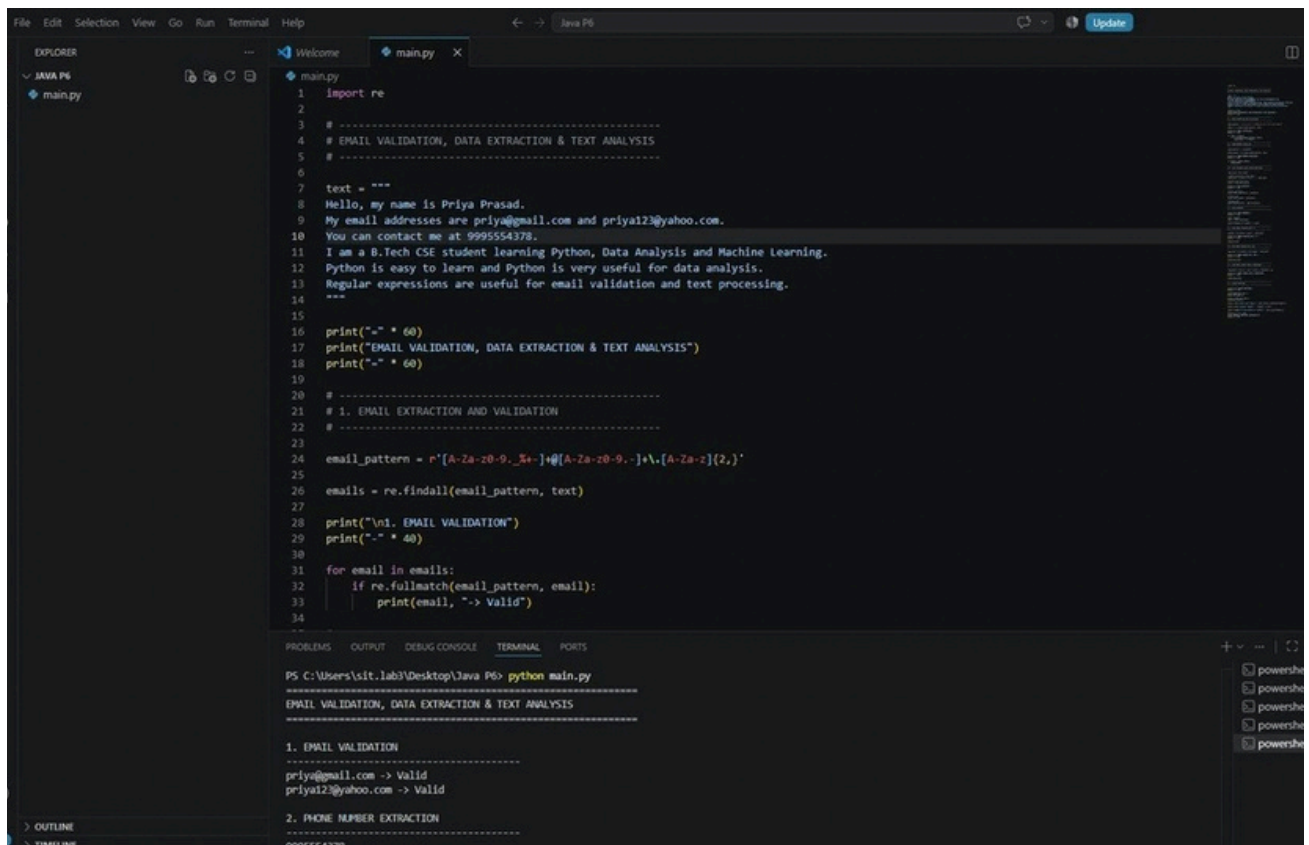
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Conclusion : In this experiment, we successfully studied arrays and loops in JavaScript. We implemented different array methods to add, remove, and manipulate elements. We also used `map()`, `filter()`, `reduce()`, and `forEach()` for array processing. The case study helped us find the maximum and minimum values from an array of numbers and understand the practical use of JavaScript array methods.

Experiment No.: 6

1. **Title:** Use string functions and regex for email validation, data extraction, and text analysis.
2. **Software/Tools Required:** Visual Studio Code (VS Code), web browser, HTML, JavaScript.
3. **Theory:** String functions and Regular Expressions (Regex) are used in JavaScript to process and analyze text. String functions such as `toLowerCase()`, `toUpperCase()`, `includes()`, `split()`, `trim()`, and `replace()` help manipulate and search text. Regular Expressions are patterns used to find specific text formats, such as validating an email address, extracting numbers or words, and identifying patterns in a sentence. In a basic HTML, CSS, and JavaScript application, HTML provides the input fields and structure, CSS provides styling, and JavaScript uses string functions and Regex to validate emails, extract required data, and perform simple text analysis.
4. **Output :**



```
1 import re
2
3 # -----
4 # EMAIL VALIDATION, DATA EXTRACTION & TEXT ANALYSIS
5 # -----
6
7 text = """
8 Hello, my name is Priya Prasad.
9 My email addresses are priya@gmail.com and priya123@yahoo.com.
10 You can contact me at 9995554378.
11 I am a B.Tech CSE student learning Python, Data Analysis and Machine Learning.
12 Python is easy to learn and Python is very useful for data analysis.
13 Regular expressions are useful for email validation and text processing.
14 """
15
16 print("-" * 60)
17 print("EMAIL VALIDATION, DATA EXTRACTION & TEXT ANALYSIS")
18 print("-" * 60)
19
20 # -----
21 # 1. EMAIL EXTRACTION AND VALIDATION
22 # -----
23
24 email_pattern = r'[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}'
25
26 emails = re.findall(email_pattern, text)
27
28 print("\n1. EMAIL VALIDATION")
29 print("-" * 40)
30
31 for email in emails:
32     if re.fullmatch(email_pattern, email):
33         print(email, "-> Valid")
34
35
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80
81
82
83
84
85
86
87
88
89
90
91
92
93
94
95
96
97
98
99
100
```

PS C:\Users\sit.labi\Desktop\Java P6> python main.py

```
=====
EMAIL VALIDATION, DATA EXTRACTION & TEXT ANALYSIS
=====

1. EMAIL VALIDATION
-----
priya@gmail.com -> Valid
priya123@yahoo.com -> Valid

2. PHONE NUMBER EXTRACTION
-----
9995554378
```

Conclusion: String functions and regular expressions (Regex) are important features of JavaScript that make text processing easier and more efficient. String functions can be used to search, compare, split, replace, trim, and change text, while Regex helps identify and work with specific patterns in the given data. These techniques can be applied to email validation, data extraction, keyword searching, and basic text analysis. When combined with HTML for creating the webpage and CSS for styling, JavaScript provides the required functionality to make the application interactive and user-friendly. Overall, learning string functions and Regex provides a strong foundation for handling text-based data and developing practical web applications.



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Case Study : 6

1. Title: To Perform Basic String Operations, Reverse a String, and Count the Number of Vowels in a Paragraph Using JavaScript.

2. Software/Tools Required: HTML, VS Code.

3. Theory: String operations in JavaScript are used to manipulate and process text. In this experiment, JavaScript string methods such as `length`, `toUpperCase()`, `toLowerCase()`, `split()`, and `reverse()` are used to perform basic string operations. A for loop and the `includes()` method are used to count the number of vowels in a paragraph. HTML is used to create the webpage, CSS is used for styling, and JavaScript is used to perform operations and display the results dynamically.

4. Output:

String Operations in JavaScript

Enter a Paragraph:

I am in Symbiosis
I am in third year
I am a student
My name is Priya Prasad

Perform Operations

Results

Original String:
I am in Symbiosis I am in third year I am a student My name is Priya Prasad

String Length: 75

Uppercase:
I AM IN SYMBIOSIS I AM IN THIRD YEAR I AM A STUDENT MY NAME IS PRIYA PRASAD

Lowercase:
i am in symbiosis i am in third year i am a student my name is priya prasad



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Conclusion : In this experiment, we successfully performed basic string operations using HTML, CSS, and JavaScript. We learned how to reverse a string and count the number of vowels in a paragraph using loops and string methods. This experiment helped us understand string manipulation and dynamic result display in JavaScript.

Experiment No.: 7

1. Title : To-Do List Application Using DOM Traversal and DOM Manipulation

2. Software/Tools Required: Visual Studio Code (VS Code), web browser, HTML, JavaScript.

3. Theory: A To-Do List Application is a simple web application used to manage and organize tasks. It allows users to add new tasks, edit existing tasks, and delete completed or unwanted tasks.

The application uses HTML, CSS, and JavaScript. HTML provides the structure of the webpage, CSS is used for designing and styling the application, and JavaScript provides the interactive functionality.

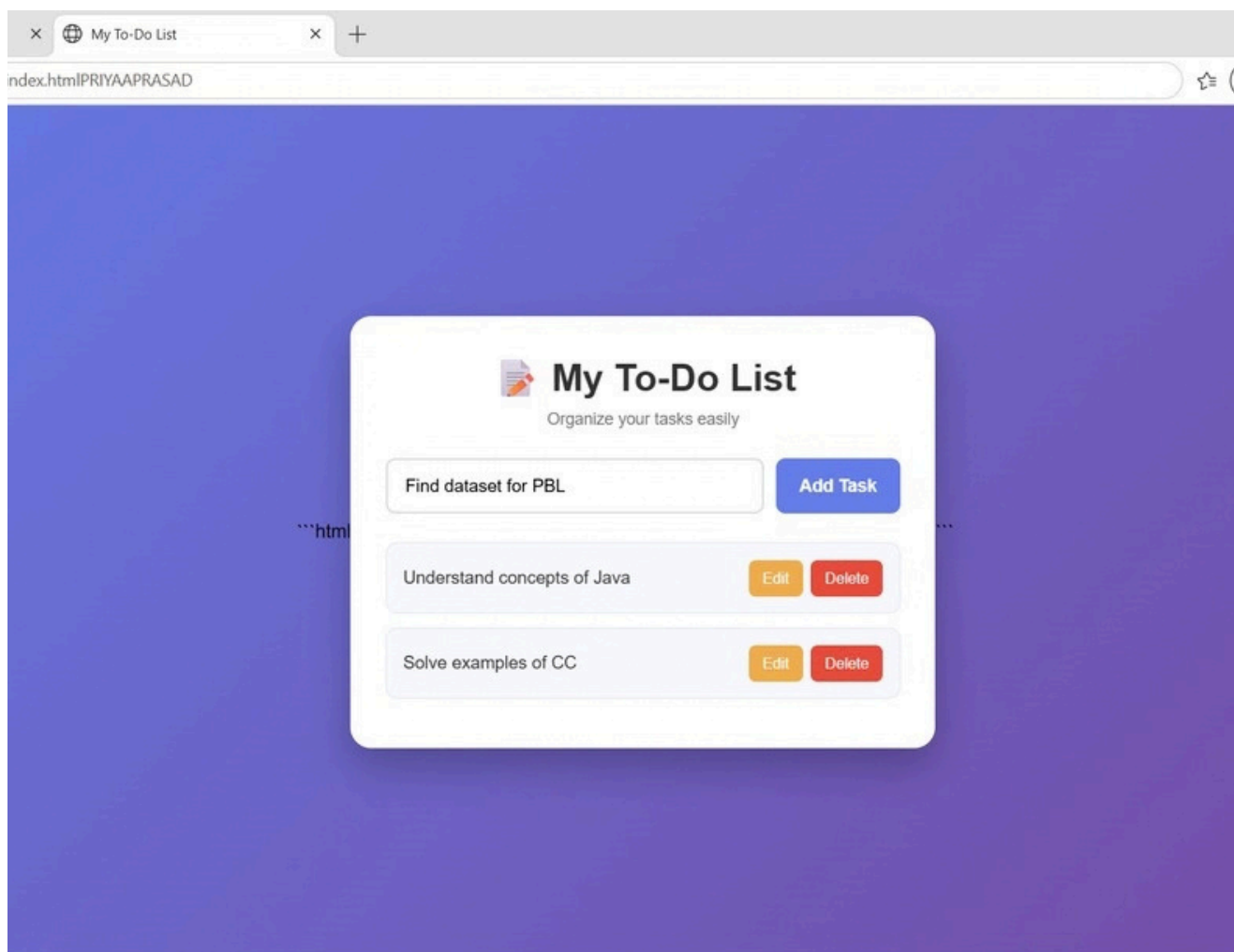
The main concept used in this application is the Document Object Model (DOM). The DOM represents the HTML document as a tree of elements. JavaScript can access these elements and perform operations on them.

DOM Traversal is the process of moving through the DOM tree to locate specific elements. Methods such as `parentElement` and `querySelector()` are used to find the required task elements.

DOM Manipulation allows JavaScript to dynamically create, modify, and remove elements from the webpage. In the To-Do List, new task elements are created using `createElement()`, added using `appendChild()`, modified using `textContent`, and removed using `removeChild()`.

Event handling is also used to respond to user actions such as clicking the Add, Edit, and Delete buttons. Thus, the application demonstrates how DOM traversal and manipulation can be used to create an interactive and dynamic webpage.

4. Output :



5. Conclusion: The To-Do List application successfully demonstrates DOM traversal and DOM manipulation using JavaScript. Users can easily add, edit, and delete tasks dynamically without refreshing the webpage. The project provides a practical understanding of how JavaScript interacts with HTML elements and helps in developing interactive and user-friendly web applications.



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Case Study : 7

1. Title: To Demonstrate Event Handling in JavaScript Using Form Elements and Input Validation.

2. Software/Tools Required: HTML, VS Code.

3. Theory: Event handling in JavaScript is used to respond to user actions on a webpage. In this experiment, HTML is used to create a student registration form, CSS is used for styling, and JavaScript is used to access and manipulate form elements. Different events such as focus, change, and submit are implemented. JavaScript validates user input, checks for empty fields, and verifies the email format before displaying the result.

4. Output:

The screenshot shows a web browser window with multiple tabs. The active tab is titled 'JavaScript Event Handling'. The address bar shows the URL '127.0.0.1:5500/index.html'. The main content area displays a 'Student Registration Form' with the following fields:

- Name:** A text input field containing 'Priya prasad'.
- Email:** A text input field containing 'priyaprasad@gmail.com'.
- Course:** A dropdown menu with 'Computer Science' selected.
- Submit:** A green button labeled 'Submit'.

Below the form, a green message states 'Form submitted successfully!'.



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Conclusion : In this experiment, we successfully implemented event handling in JavaScript using a student registration form. We learned how to access form elements, handle focus, change, and submit events, and validate user inputs. This experiment helped us understand how JavaScript makes webpages interactive and user-friendly.



Experiment No.: 8

1. **Title :** Access and validate form fields using events; design a gym admission form with live input checks.

2. **Software/Tools Required:** Visual Studio Code (VS Code), web browser, HTML, JavaScript.

3. **Theory:** Form validation is the process of checking whether the information entered by a user is correct and complete before submitting a form. In this program, a Gym Admission Form is created using HTML, styled using CSS, and validated using JavaScript.

JavaScript accesses the form fields using DOM methods such as `document.getElementById()`. Events are used to detect user actions. The input event performs live validation while the user is typing, the change event checks changes in selections, and the submit event validates the complete form before submission.

The program checks important fields such as name, email, phone number, age, gender, and membership plan. If the entered data is incorrect, an appropriate error message is displayed. Regular expressions are used to check patterns such as email and phone numbers. The `preventDefault()` method is used to stop form submission when the entered information is invalid.

Thus, the program demonstrates DOM manipulation, event handling, and form validation to create an interactive and user-friendly gym admission form.

4. Output :

The screenshot shows a web browser window with the address bar displaying "127.0.0.1:5500/index.html,priyaprasad.24070521179". The page contains a "Gym Admission Form" with the subtitle "Fill in your details to join the gym". The form fields are filled with the following information:

- Full Name:** Priya Prasad
- Email:** priya.prasad.batch2024@sitnagpur.siu.edu.in
- Phone Number:** 1234567892
- Age:** 20
- Gender:** ☒ Female
- Membership Plan:** Yearly

Below the form fields is a black "Join Gym" button. At the bottom of the form, a green message reads "Admission successful! Welcome to the gym."

The screenshot shows a web browser window with the address bar displaying "127.0.0.1:5500/index.html,priyaprasad.24070521179". The page contains a "Gym Admission Form" with the subtitle "Fill in your details to join the gym". The form fields are filled with the following information:

- Full Name:** Priya Prasad
- Email:** priya.prasad.batch2024@sitnagpur.siu.edu.in
- Phone Number:** 1234567892
- Age:** 11
- Gender:** ☒ Female
- Membership Plan:** Yearly

Below the form fields is a black "Join Gym" button. A red error message is displayed below the age field: "Enter a valid age between 13 and 100." At the bottom of the form, a green message reads "Admission successful! Welcome to the gym."

5.Conclusion: The Gym Admission Form successfully demonstrates the use of HTML, CSS, and JavaScript for creating and validating a web form. JavaScript events provide live input validation, display appropriate error messages, and prevent invalid form submission. This makes the form more interactive, accurate, and user-friendly.



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Case Study : 8

1. Title: Online Job Application Form Using HTML, CSS, and JavaScript

2. Software/Tools Required: HTML, VS Code.

3. Theory: The Online Job Application Form is developed using HTML, CSS, and JavaScript. HTML is used to create the form structure and input fields such as name, email, phone number, qualification, and job role. CSS is used to style the form and improve its appearance. JavaScript is used to validate user inputs, check empty fields, validate email and phone numbers, and handle the form submission event. DOM manipulation is used to display success and error messages dynamically.

4. Output:

Online Job Application Form

Full Name:
Priya Prasad

Email:
priya.prasad.batch2024@sitnagpur.siu.edu.in

Phone Number:
9876234567

Qualification:
B.Tech

Job Role:
Software Developer

Apply Now

Application submitted successfully!



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Conclusion : In this case study, we successfully created an Online Job Application Form using HTML, CSS, and JavaScript. We implemented form validation, event handling, and dynamic message display. This activity helped us understand how JavaScript can be used to access form elements, validate user input, and improve the functionality of web forms.

Experiment No.: 9

1. **Title :** Implementation of localStorage and sessionStorage to Save User Preferences

2. **Software/Tools Required:** Visual Studio Code (VS Code), web browser, HTML, JavaScript.

3. **Theory:** Web Storage is a feature of JavaScript that allows websites to store data in the user's browser. It provides two main storage methods: localStorage and sessionStorage.

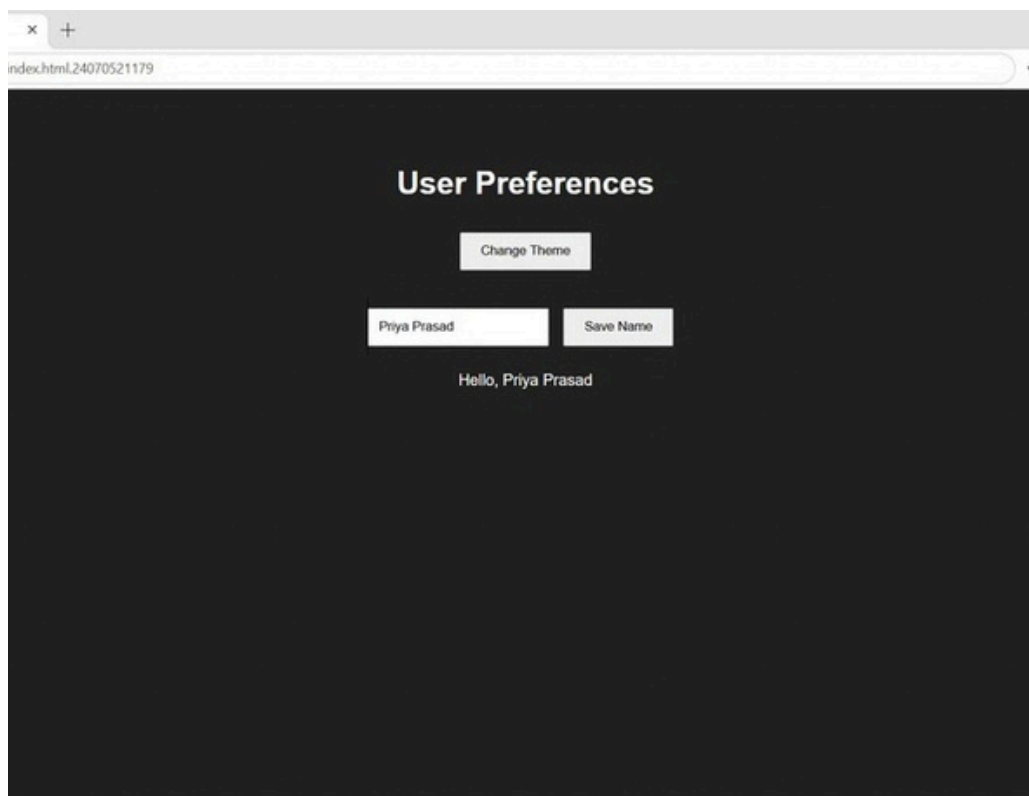
localStorage is used to store data that remains available even after closing and reopening the browser. In this practical, localStorage is used to save the user's theme preference, such as light mode or dark mode.

sessionStorage is used to store data for a particular browser session. In this practical, it is used to save the user's name. The stored data remains available while the session continues and is generally removed when the browser tab or session ends.

JavaScript methods such as `setItem()` and `getItem()` are used to save and retrieve data from storage. The `classList.toggle()` method is used to change the theme dynamically.

This practical demonstrates how web applications can remember user preferences and provide a personalized user experience.

4. Output :



5.Conclusion: In this practical, we successfully implemented preferences. LocalStorage was used to store the theme settings, while sessionStorage was used to store the username. The theme was changed dynamically using JavaScript, and stored preferences were retrieved when the page was reloaded. This practical helped us understand browser storage and its use in web development.

Experiment No.: 10

1. **Title:** Loading and Displaying JSON Data Using Fetch API and jQuery getJSON() with Dynamic Table Rendering

2. **Software/Tools Required:** Visual Studio Code (VS Code), web browser, HTML, JavaScript.

3. **Theory:** JSON (JavaScript Object Notation) is a lightweight format used to store and exchange data between a server and a web application. It represents data in a structured format that is easy for humans to read and machines to process.

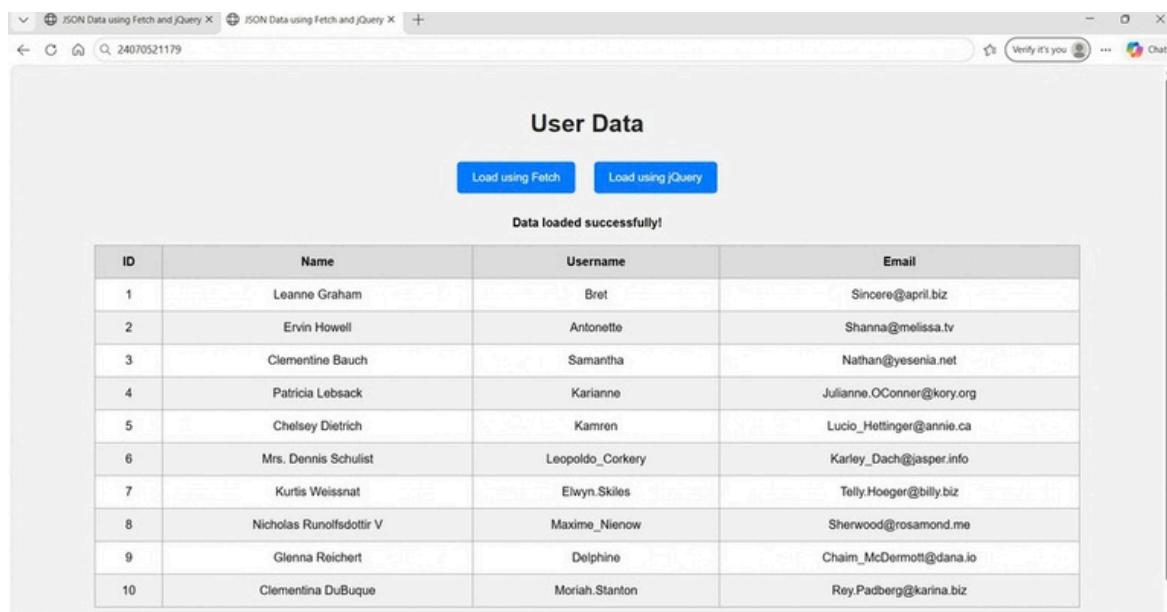
The Fetch API is a modern JavaScript method used to request data from an external API. It retrieves JSON data using the `fetch()` function and converts the response into a JavaScript object using `response.json()`.

jQuery `getJSON()` is a jQuery method used to load JSON data from a server. It simplifies the process of requesting and handling JSON responses.

In this practical, JSON data is fetched from an API using both Fetch API and jQuery `getJSON()`. The retrieved user information, such as ID, name, username, and email, is displayed dynamically in an HTML table using JavaScript. The table is updated by creating rows and inserting the data into the table body.

This practical demonstrates how web applications retrieve external data, process JSON responses, and display information dynamically without manually entering each table row.

Output :



The screenshot shows a web browser window with two tabs. The active tab is titled 'JSON Data using Fetch and jQuery X'. The address bar shows a local file path. The page content includes a heading 'User Data', two blue buttons labeled 'Load using Fetch' and 'Load using jQuery', and a message 'Data loaded successfully!'. Below this is a table with 10 rows of user data.

ID	Name	Username	Email
1	Leanne Graham	Bret	Sincere@april.biz
2	Ervin Howell	Antonette	Shanna@melissa.tv
3	Clementine Bauch	Samantha	Nathan@yesenia.net
4	Patricia Lebsack	Karianne	Julianne.OConner@kory.org
5	Chelsey Dietrich	Kamren	Lucio_Hettinger@annie.ca
6	Mrs. Dennis Schulist	Leopoldo_Corkery	Karley_Dach@jasper.info
7	Kurtis Weissnat	Elwyn Skiles	Telly.Hoeger@billy.biz
8	Nicholas Runolfsdottir V	Maxime_Nienow	Sherwood@rosamond.me
9	Glenna Reichert	Delphine	Chaim_McDermott@dana.io
10	Clementina DuBuque	Moriah Stanton	Rey.Padberg@karina.biz

Conclusion: In this practical, we learned how to fetch and load JSON data using the Fetch API and jQuery getJSON(). The retrieved data was displayed dynamically in an HTML table using JavaScript. This practical helped us understand API data handling, JSON parsing, and dynamic content rendering in web development.